Calum Smith

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EDUCATION

University of Warwick (Classification Pending)
MSc Statistics with Data Science Sep 2025 – Sep 2026

Modules: Monte Carlo Methods, Statistical Inference, Bayesian Forecasting, Machine Learning

Durham University, St. Chad's College
*MSc Finance — **Distinction**; Business School Scholarship* Oct 2019 – Sep 2020

University of Manchester
*BSc (Hons) in Mathematics — **2:1*** Sep 2015 – Jul 2018

RESEARCH & AI SAFETY

A Practical Block-Sampling Particle Filter: Amortising Online Inference 2026
Warwick MSc dissertation, supervised by Prof. Adam Johansen

- Extended block-sampling particle filters beyond linear Gaussian models: a conditional VAE learns the proposal offline, so each online step costs one forward pass, and a single network serves as both proposal and artificial distribution.
- Reduced resampling in a linear Gaussian model from ~ 113 to ~ 20 steps per 500-step run as block length grew from 1 to 6, versus no reduction for an independently trained network; gains were smaller on harder models and filtered-state accuracy was unchanged.
- Showed the ELBO does not control importance-weight variance (a χ^2-discrepancy that can be infinite while the ELBO is small), so proposals were validated through filter diagnostics, not the training loss.

AI Safety Fellowship, Technical Track | *Warwick EA & AI Societies* Autumn 2025

- Participated in the technical track; presented on variational autoencoders, the model class central to my dissertation.

Deep Reinforcement Learning Coursework | *Advanced Topics in Statistics & Probability* 2026

- Implemented and compared Q-learning and SARSA on a GridWorld, and DQN against REINFORCE on CartPole, including a target-network ablation.

OTHER QUANTITATIVE PROJECTS

Time Series Forecasting & State-Space Modelling | *Modelling Lead, Statistical Consulting (Warwick)* 2026

- Led modelling (SARIMA, Kalman state-space, BSTS) for a 6-person consultancy project forecasting UK temperatures.
- Rejected a better-fitting BSTS model after diagnosing posterior-variance behaviour that made long-horizon intervals unusable; recommended a more stable model, cross-checked on a second dataset.

Advanced Monte Carlo & MCMC Methods | *R, Python (Warwick)* 2025

- Implemented MALA, random-walk Metropolis–Hastings and Gibbs samplers, including for the 2D Ising model.

WORK EXPERIENCE

Self-Directed Mathematical Study *(concurrent: Clinical Support Worker, NHS, 10 months)* Jan 2024 – Sep 2025

- Rebuilt foundations in statistical inference and Bayesian statistics ahead of the Warwick MSc.

Securing Energy for Europe (SEFE), Manchester & London
Quantitative Analyst, *Gas Pricing* (Jul 2022 – Dec 2023); *Flexible Portfolio Analyst* (May 2021 – Jul 2022)

- Produced and quality-controlled daily forward gas curves (NBP, TTF), the primary pricing reference for B2B supply contracts in the UK and Netherlands.
- Maintained the pricing framework (intraday volatility, risk adders) and supported automated adder generation via Trayport through structured testing.
- Produced monthly Cost of Power reports and reconciled data supporting £100m+ monthly billing on a flexible gas portfolio.

TECHNICAL SKILLS

- **Programming:** Python (NumPy, Pandas, scikit-learn, PyTorch), R (`KFAS`, `bsts`, `forecast`), Git, LaTeX
- **Methods:** Monte Carlo/MCMC/SMC; Bayesian inference; state-space models; VAEs